

GEO-02 Teacher Diagnostic Key

Teacher material - answers and diagnostics

Teacher Diagnostic / Answer Key

Assimilation attempts

- GEO02-A01: Adjacent obtuse angle = $180-73 = 107$ degrees.
- GEO02-A02: Basic rotation is $360/15=24$ degrees; $72/24=3$, so yes.

Recognition / first-line lab

- GEO02-R01: Transfer 64 by corresponding angles, then adjacent = 116.
- GEO02-R02: By the converse, the two lines are parallel.
- GEO02-R03: Fourth angle = $360-(84+96+111)=69$.
- GEO02-R04: Use triangle closure separately on each side of the diagonal.
- GEO02-R05: Exterior turn = $360/18=20$, so interior = 160.
- GEO02-R06: Write $k=360/(180-\theta)$ and enforce integer $k \geq 3$.
- GEO02-R07: $n(n-3)/2 = 8*5/2=20$.
- GEO02-R08: Do not use symmetry; identify an additional equality/congruence that would prove it.
- GEO02-R09: Use regular-polygon rotational/gap symmetry; do not test every pair.
- GEO02-R10: Test coordinate/vector placement because length+direction data are numerical.
- GEO02-R11: Choose bases/legs and impose $|p-q| < d < p+q$ before counting.
- GEO02-R12: Equiangular alone does not guarantee the claimed reflection symmetry.

Practice / transfer bank

- GEO02-P01: 72,108
- GEO02-P02: 47,133
- GEO02-P03: 77
- GEO02-P04: 132
- GEO02-P05: 15
- GEO02-P06: 15
- GEO02-P07: No
- GEO02-P08: 44
- GEO02-P09: 11
- GEO02-P10: Yes
- GEO02-P11: No
- GEO02-P12: No
- GEO02-P13: Parallel
- GEO02-P14: Yes
- GEO02-P15: No; degenerate
- GEO02-P16: Coordinate/vector

Mixed mastery

- GEO02-M01: Lines are parallel by corresponding-angle converse; adjacent angle 113.
- GEO02-M02: 79 degrees.
- GEO02-M03: $B=77$, $D=91$, sum 168.
- GEO02-M04: Exterior 15, hence $n=24$.
- GEO02-M05: $14*11/2=77$.

- GEO02-M06: Yes. AC is common, so $AB=AD$, $BC=CD$ and $AC=AC$ give triangles ABC and ADC congruent by SSS. Equivalently A and C are both equidistant from B and D, so AC is the perpendicular bisector of BD and reflection across AC interchanges B and D.
- GEO02-M07: Basic rotation 20; 80 is 4 turns, so yes.
- GEO02-M08: Angle-labelled: synthetic closure first. Length+direction: coordinate/vector representation first.
- GEO02-M09: Parallelism was assumed from appearance. Replace by a given parallel relation or use a converse from a proved angle equality.
- GEO02-M10: Exterior angle is $180-\theta$ and $n=360/\text{exterior}$; polygon requires integer $n \geq 3$.
- GEO02-M11: Rotational symmetry groups diagonals by direction. Perpendicularity requires a 90-degree direction difference, so count compatible direction classes (or equivalent endpoint-gap classes) rather than checking every pair individually.
- GEO02-M12: No. Equality forces zero height; strict feasibility is required.

Historical anchor audit

- IOQM-2025-Q13 = 60 (independently recomputed; agrees with FINAL_OFFICIAL key).
- IOQM-2024-Q04 = 70 (independently recomputed; agrees with HBCSE key).
- IOQM-2023-Q06 = 16 (independently recomputed; agrees with embedded key).
- IOQM-2023-Q24 = 31 (independently recomputed; agrees with embedded key).
- IOQM-2023-Q25 = 28 (independently recomputed; agrees with embedded key).